Mathematics Standard
CBSE Examination Paper – 2022
(Term-2)

With Topper's Answer

Time : 2:00 Hours **Max. Marks : 40**

General Instructions:

Read the following instructions very carefully and strictly follow them :

(i) *This question paper contains **14** questions. **All** questions are compulsory.*

(ii) *This question paper is divided into **three** sections **Sections A, B** and **C.***

(iii) ***Section A** comprises of **6** questions (Q.no. **1** to **6**) of **2** marks each. Internal choice has been provided in **two** questions.*

(iv) ***Section B** comprises of **4** questions (Q.no. **7** to **10**) of **3** marks each. Internal choice has been provided in **one** question.*

(v) ***Section C** comprises of **4** questions (Q.no. **11** to **14**) of **4** marks each. Internal choice has been provided in **one** question. It also contains two case study based questions.*

(vi) *Use of calculator is **not** permitted.*

Set-30/3/3

SECTION – A

Question numbers 1 to 6 carry 2 marks each.

1. In Figure 1, PQ and PR are tangents to the circle centred at O. If ∠OPR = 45°, then prove that ORPQ is a square. **2**

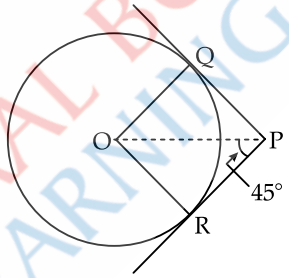


Figure 1

Ans.

To prove :- ORPQ is a square i.e. ∠O = ∠Q = ∠P = ∠R = 90° and OQ = QP = PR = OR

Proof :- ∠OQP = 90°] Tangent is perpendicular to —(1)
∠ORP = 90°] the radius at the point of contact.

Now, In △ORP and △OQP
OP = OP (common)
OR = OQ (Radii of same circle)
PR = PQ (Tangents from an external point to a circle are equal)
Therefore, △ORP ≅ △OQP by SSS rule
∴ ∠OPR = ∠OPQ (By CPCT)

⇒ ∠OPR = ∠OPQ = 45°
⇒ ∠OPR + ∠OPQ = 90°
⇒ ∠QPR = 90° —(2)

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In quad. OQPR

$\angle OQP + \angle QPR + \angle PRO + \angle ROQ = 360°$ (Angle Sum Property)

$90 + 90 + 90 + \angle ROQ = 360°$

$\angle ROQ = 360 - 270$

$\angle ROQ = 90°$ — (3)

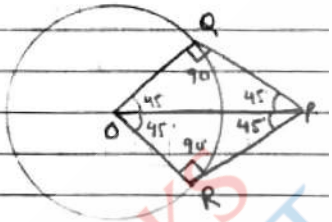
Also $OR = PR$ — (4)

(As $\angle ROP = \angle RPO = 45$) (Isosceles triangle property)

Also $QP = PR$ (Tangents from an external point to a circle are equal) — (5)

$\Rightarrow$ Using (4) and (5)

$OQ = PQ = PR = OR$ — (6)



Using (1), (2), (3), (6)

ORPQ is a square.

Hence proved.

2. Find the mode of the given frequency distribution : **2**

Class	Frequency
15 – 25	6
25 – 35	11
35 – 45	22
45 – 55	23
55 – 65	14
65 – 75	5

Ans.

Class	Frequency	
15-25	6	
25-35	11	
35-45	f_0	22
45-55	f_1	23 → Modal class
55-65	f_2	14
65-75		5

$$\text{Mode} = l + \left\{\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right\} \times h$$

$$= 45 + \left\{\frac{23 - 22}{46 - 22 - 14}\right\} \times 10$$

$$= 45 + \left\{\frac{1}{10} \times 10\right\}$$

$$= 45 + 1 = \boxed{46}$$

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3. Find the common difference '*d*' of an AP whose first term is 10 and the sum of the first 14 terms is 1505. **2**

Ans.

$S_{14} = \frac{n}{2}(2a+(n-1)d)$

$1505 = \frac{14}{2}(2(10)+13d)$

$1505 = 7(20+13d)$

$\frac{1505}{7} = 20+13d$

$215 = 20+13d$

$\frac{195}{13} = d$

$d = 15$

4. For what value of '*n*', are the nth terms of the APs : 9, 7, 5, ... and 15, 12, 9, ... the same ? **2**

Ans.

Quest 9, 7, 5 ——— and 15, 12, 9 ———

$a_n = a+(n-1)d$ $\quad a'_n = a'+(n-1)d'$

$= 9+(n-1)-2$ $\quad = 15+(n-1)-3$

$= 9-2n+2$ $\quad = 15-3n+3$

$= 11-2n$ $\quad = 18-3n$

since $a_n = a'_n$

$11-2n = 18-3n$

$11-18 = -3n+2n$

$-7 = -n$

$\boxed{n = 7}$

5. (a) Solve the quadratic equation for x :

$x^2 - 2ax - (4b^2 - a^2) = 0$ **2**

OR

(b) If the quadratic equation

$$(1 + a^2)\,x^2 + 2abx + (b^2 - c^2) = 0$$

has equal and real roots, then prove that :

$$b^2 = c^2\,(1 + a^2)$$ **2**

Ans.(a)

$x^2 - 2ax - (4b^2 - a^2) = 0$

$b^2 - 4ac = 4a^2 - 4[-(4b^2 - a^2)](1)$

$= 4a^2 - 4[-4b^2 + a^2]$

$= 4a^2 + 16b^2 - 4a^2$

$= 16b^2$

$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{2a \pm \sqrt{16b^2}}{2(1)} = \frac{2a \pm 4b}{2}$

$\Rightarrow x = \frac{2a+4b}{2}$ $\quad x = \frac{2a-4b}{2}$

$x = a+2b$ $\quad x = a-2b$

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6. (a) 150 spherical marbles, each of diameter 1·4 cm, are dropped in a cylindrical vessel of diameter 7 cm containing some water, and are completely immersed in water. Find the rise in the level of water in the cylindrical vessel. **2**

OR

(b) Three cubes of side 6 cm each, are joined as shown in Figure 2. Find the total surface area of the resulting cuboid. **2**

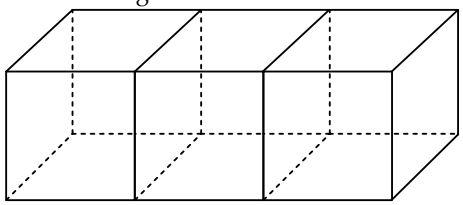


Figure 2

Ans.(b)

Length of cuboid = 18cm
Breadth of cuboid = 6cm
Height of cuboid = 6cm

TSA of the cuboid = $2(lb+bh+hl)$
$= 2(18\times 6+6\times 6+6\times 18)$
$= 2(108+36+108)$
$= 2(252)$
$= 504\ cm^2$

SECTION B

Question numbers 7 to 10 carry 3 marks each.

7. Two boats are sailing in the sea 80 m apart from each other towards a cliff AB. The angles of depression of the boats from the top of the cliff are 30° and 45° respectively, as shown in Figure 3. Find the height of the cliff. **3**

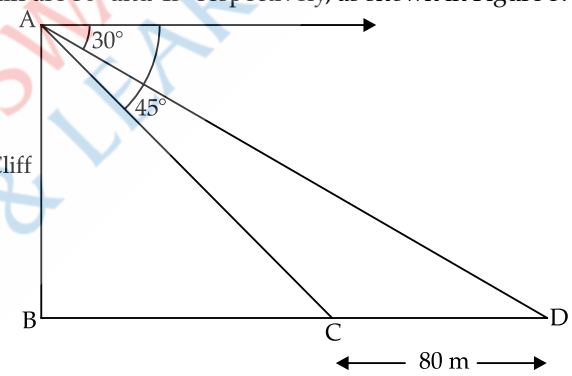
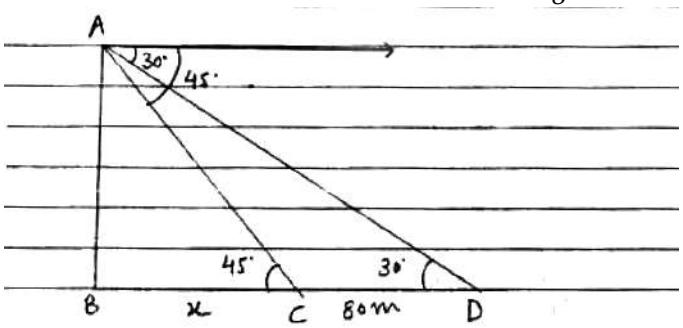


Figure 3

Ans.



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In ΔABC

$\tan 45° = \frac{P}{B} = \frac{AB}{x}$

$1 = \frac{AB}{x}$

$AB = x$

In ΔABD

$\tan 30° = \frac{P}{B} = \frac{AB}{80+x} \Rightarrow \frac{1}{\sqrt{3}} = \frac{AB}{80+x}$

$\frac{80+x}{\sqrt{3}} = AB$

$\frac{80+x}{\sqrt{3}} = x$

$80 + x = \sqrt{3}x$

$80 = \sqrt{3}x - x$

$80 = x(\sqrt{3}-1)$

$\frac{80}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} = x$

$\frac{80(\sqrt{3}+1)}{(\sqrt{3})^2 - 1^2} = x$

$\frac{80(\sqrt{3}+1)}{3-1} = x$

$\frac{80(\sqrt{3}+1)}{2} = x$

$x = 40(\sqrt{3}+1)$ m

$\Rightarrow AB = 40(\sqrt{3}+1)$ m or 109.2 m

8. For the following frequency distribution, find the median : **3**

Class	Frequency
1400 – 1550	6
1550 – 1700	13
1700 – 1850	25
1850 – 2000	10

Ans.

Class	Frequency	C.F.
1400–1550	6	6
1550–1700	13	19
1700–1850	25	44 → Median class
1850–2000	10	54 = n

$\text{Median} = l + \left(\frac{n - CF}{\frac{2}{f}}\right) \times h$

$= 1700 + \left(\frac{27 - 19}{25}\right) \times 150$

$= 1700 + \left(\frac{8}{25}\right) \times 150^{6}$

$= 1700 + 48$

$= 1748$

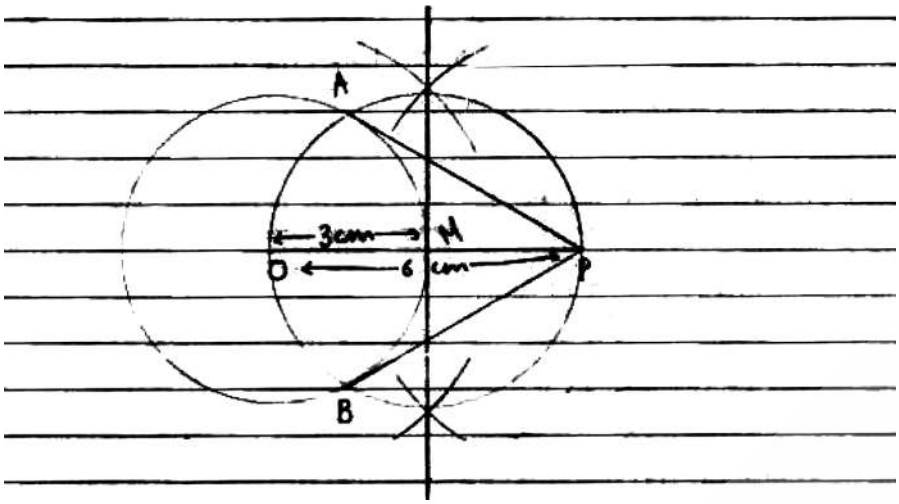
9. (a) Draw a line segment AB of length 8 cm and locate a point P on AB such that AP : PB = 1 : 5. **3**

OR

(b) Draw a circle of radius 3 cm. From a point P lying outside the circle at a distance of 6 cm from its centre, construct two tangents PA and PB to the circle. **3**

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Ans.(b)



Steps of construction →

1. Draw a circle with O as centre and radius 3cm.
2. Take a point P outside the circle at a distance of 6cm from its centre O.
3. Join OP.
4. Construct the perpendicular bisector of OP
5. Name the mid-point of OP as M.
6. Draw a circle with H as centre and radius OM.
7. The second circle intersects the first circle at two points Name them A and P
8. Connect PA and PB.
9. PA and PB are the required tangents.

10. Find the mean of the following frequency distribution : **3**

Class	Frequency
0 – 10	12
10 – 20	18
20 – 30	27
30 – 40	20
40 – 50	17
50 – 60	6

Ans.

Class	Frequency (fi)	xi	fixi
0–10	12	5	60
10–20	18	15	270
20–30	27	25	675
30–40	20	35	700
40–50	17	45	765
50–60	6	55	330
	100		2800

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Mean $= \frac{\Sigma f_i x_i}{\Sigma f_i}$

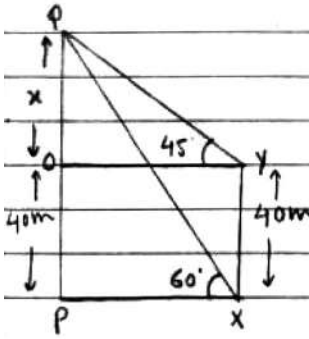
$\bar{x} = \frac{2800}{100} = 28$

SECTION C

Question numbers 11 to 14 carry 4 marks each.

11. The angle of elevation of the top Q of a vertical tower PQ from a point X on the ground is 60°. From a point Y, 40 m vertically above X, the angle of elevation of the top Q of tower PQ is 45°. Find the height of the tower PQ and the distance PX. [Use $\sqrt{3} = 1.73$] **4**

Ans.



In $\triangle QOY$

$\tan 45 = \frac{P}{B} = \frac{QO}{OY} = \frac{x}{OY}$

$1 = \frac{x}{OY}$

$OY = x \Rightarrow PX = x$

In $\triangle QPX$

$\tan 60 = \frac{P}{B} = \frac{QP}{PX} = \frac{40+x}{x}$

$\sqrt{3} = \frac{40+x}{x}$

$\sqrt{3}x = 40 + x$

$\sqrt{3}x - x = 40$

$x(\sqrt{3}-1) = 40$

$x = \frac{40}{(\sqrt{3}-1)} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} = \frac{40(\sqrt{3}+1)}{(\sqrt{3})^2 - 1^2} = \frac{40(\sqrt{3}+1)}{3-1}$

$x = \frac{40(\sqrt{3}+1)}{2}$

$x = 20(\sqrt{3}+1)$ m

$x = 54.6$ m

$\Rightarrow PX = 54.6$ m

$PQ = 40 + x$

$= 40 + 54.6$

$= 94.6$ m

12. (a) In Figure 4, two circles with centres at O and O of radii $2r$ and r respectively, touch each other internally at A. A chord AB of the bigger circle meets the smaller circle at C. Show that C bisects AB. **4**

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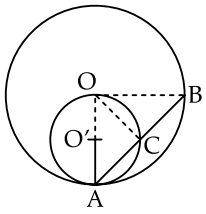


Figure 4

OR

(b) In Figure 5, O is centre of a circle of radius 5 cm. PA and BC are tangents to the circle at A and B respectively. If OP = 13 cm, then find the length of tangents PA and BC. **4**

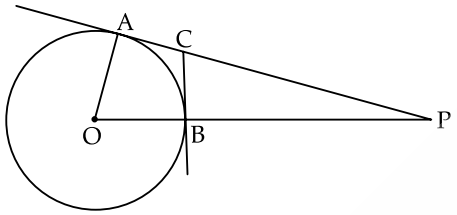
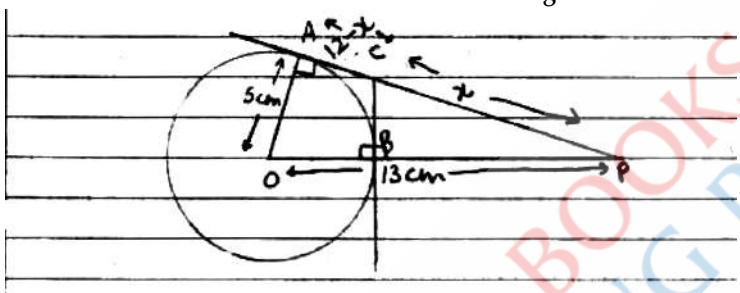


Figure 5

Ans.



$\angle OAP = 90°$ (Tangent is perpendicular to radius at the point of contact)

Using pythagoras theorem in $\triangle OAP$

$H^2 = B^2 + P^2$

$OP^2 = AP^2 + OA^2 \Rightarrow AP^2 = 144$

$13^2 = AP^2 + 5^2$ $\quad AP = \sqrt{144} = 12\text{cm}$

$169 = AP^2 + 25$

$169 - 25 = AP^2$

Let PC be x and AC be 12-x

AC = BC = 12-x (Tangents from point c to the circle are equal)

Also OP = 13cm

OB = 5cm

BP = OP - OB

BP = 8cm

$\angle OBC = 90°$ (Tangent is perpendicular to the radius at the POC)

$\Rightarrow \angle CBP = 90°$ (Linear pair)

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Using pythagoras theorem in Δ CBP

$H^2 = B^2 + P^2$

$CP^2 = BP^2 + BC^2$

$x^2 = 8^2 + (12-x)^2$

$x^2 = 64 + 144 + x^2 - 24x$

$24x = 64 + 144$

$24x = 208$

$x = \frac{208}{24} = \frac{104}{12} = \frac{52}{6} = \frac{26}{3}$

$BC = 12 - x$

$BC = 12 - \frac{26}{3}$

$BC = \frac{36-26}{3} = \frac{10}{3} = 3.34\ m$

Case Study – 1

13. In the picture given below, one can see a rectangular in-ground swimming pool installed by a family in their backyard. There is a concrete sidewalk around the pool of width x m. The outside edges of the sidewalk measure 7 m and 12 m. The area of the pool is 36 sq. m.

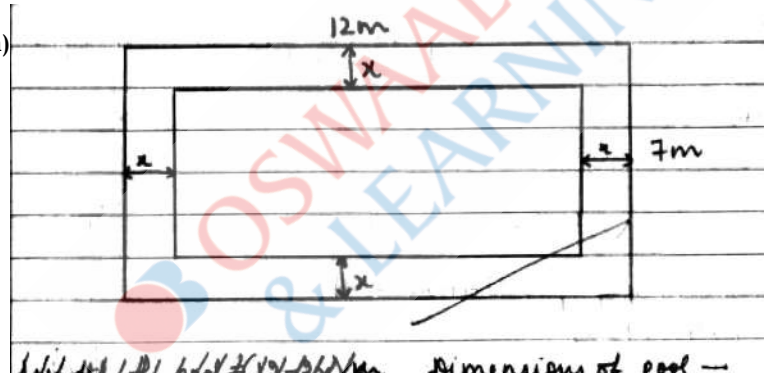


Figure 6

(a) Based on the information given above, form a quadratic equation in terms of x. **2**

(b) Find the width of the sidewalk around the pool. **2**

Ans.(a)



Length of pool = (12-2x) m

Breadth of pool = (7-2x) m

Dimensions of pool — (12-2x) and (7-2x)

Area = length x Breadth

$36 = (12-2x)(7-2x)$

$36 = (84 - 24x - 14x + 4x^2)$

$36 = 84 - 38x + 4x^2$

$36 - 84 = -38x + 4x^2$

$-48 = -38x + 4x^2$

$4x^2 - 38x + 48 = 0 \Rightarrow 2x^2 - 19x + 24 = 0$

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(b) $2x^2-19x+24=0$

$2x^2-16x-3x+24=0$

$2x(x-8)-3(x-8)=0$

$x=\frac{3}{2}$ $x=8$

$x=8$ will be neglected because if $x=8$ the width of sidewall will become more than 16 but they are already given 7m and 12m the dimensions

$\therefore$ $x=1.5$m

Case Study – 2

14. John planned a birthday party for his younger sister with his friends. They decided to make some birthday caps by themselves and to buy a cake from a bakery shop. For these two items, they decided the following dimensions :

Cake : Cylindrical shape with diameter 24 cm and height 14 cm.

Cap : Conical shape with base circumference 44 cm and height 24 cm.



Based on the above information, answer the following questions :

(a) How many square cm paper would be used to make 4 such caps ? **2**

(b) The bakery shop sells cakes by weight (0.5 kg, 1 kg, 1.5 kg, etc.). To have the required dimensions, how much cake should they order, if 650 cm^3 equals 100 g of cake ? **2**

Ans. (a) CSA of one cap = Aquare cm paper required for one cap

$2\pi x=44$

$2\times\frac{22}{7}\times x=44$

$x=\frac{44\times7}{22\times2}=7$cm

$l^2=l^2+x^2$

$l^2=24^2+7^2$

$l^2=576+49$

$l^2=625$

$l=25$ cm

CSA of one cap $=\pi rl$

$=\frac{22}{7}\times7\times25$

$=550\ cm^2$

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CSA of four caps i.e. square cm paper required for four caps = 4×550
$= 2200\ cm^2$

(b) Volume of cake $= \pi r^2 h$ | $d = 24\ cm$, $r = \frac{d}{2} = 12\ cm$

$= \frac{22}{7} \times 12 \times 12 \times 14.2$

$= 6336\ cm^3$

$650\ cm^3 = 100\ g$ or $0.1\ kg$

$6336\ cm^3 = \frac{0.1\ kg \times 6336}{650} = \frac{6336}{6500} = \frac{95.9}{100} = 0.95\ kg$

So they should order a 1 kg cake.

■■

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Mathematics Standard
CBSE Examination Paper – 2023

With Topper's Answer

Time : 3:00 Hours **Max. Marks : 80**

General Instructions:

Read the following instructions carefully and follow them:

(i) *This question paper contains **38** questions. **All** questions are **compulsory**.*

(ii) *This question paper is divided into **FIVE** sections - **A, B, C, D** and **E.***

(iii) *In **Section A,** Question numbers **1** to **18** are multiple choice questions (MCQs) and question numbers **19** and **20** are Assertion – Reason based questions of **1** mark each.*

(iv) *In **Section B,** Question numbers **21** to **25** are very short answer (VSA) type questions, carrying **2** marks each.*

(v) *In **Section C,** Question numbers **26** to **31** are short answer (SA) type questions, carrying **3** marks each.*

(vi) *In **Section D,** Question numbers **32** to **35** are long answer (LA) type questions, carrying **5** marks each.*

(vii) *In **Section E,** Question numbers **36** to **38** are **case-study based integrated** questions carrying **4** marks each. Internal choice is provided in **2** marks questions in each case-study.*

(viii) *There is no overall choice. However, an internal choice has been provided in 2 questions in Section **B**, 2 questions in Section **C**, 2 questions in Section **D** and 3 questions of 2 marks in Section **E**.*

(ix) *Draw neat diagrams wherever required. Take $\pi = \frac{22}{7}$ wherever required, if not stated.*

(x) *Use of calculators is **NOT allowed**.*

Set--30/5/1

SECTION – A

This section comprises multiple choice questions (MCQs) of 1 mark each.

1. The number of polynomials having zeroes -3 and 5 is :

(a) only one **(b)** infinite **(c)** exactly two **(d)** at most two

Ans. (b) infinite

2. The pair of equations $ax + 2y = 9$ and $3x + by = 18$ represent parallel lines, where a, b are integers, if :

(a) $a = b$ **(b)** $3a = 2b$ **(c)** $2a = 3b$ **(d)** $ab = 6$

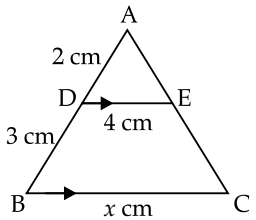
Ans. (d) $ab = 6$

3. The common difference of the A.P. whose nth term is given by $a_n = 3n + 7$, is :

(a) 7 **(b)** 3 **(c)** $3n$ **(d)** 1

Ans. (b) 3

4. In the given figure, DE || BC. The value of x is:



(a) 6 **(b)** 12.5 **(c)** 8 **(d)** 10

Ans. (d) 10

5. A quadratic equation whose roots are $(2 + \sqrt{3})$ and $(2 - \sqrt{3})$ is :

(a) $x^2 - 4x + 1 = 0$ **(b)** $x^2 + 4x + 1 = 0$ **(c)** $4x^2 - 3 = 0$ **(d)** $x^2 - 1 = 0$

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Ans. (a) $x^2 - 4x + 1 = 0$

6. If $\tan \square = \frac{5}{12}$, then the value of $\frac{\sin\theta + \cos\theta}{\sin\theta - \cos\theta}$ is :

(a) $-\frac{17}{7}$ **(b)** $\frac{17}{7}$ **(c)** $\frac{17}{13}$ **(d)** $-\frac{7}{13}$

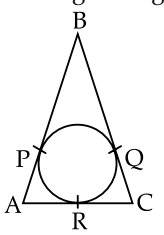
Ans. (a) $-\frac{17}{7}$

7. The distance between the points $P\left(-\frac{11}{3}, 5\right)$ and $Q\left(-\frac{2}{3}, 5\right)$ is :

(a) 6 units **(b)** 4 units **(c)** 2 units **(d)** 3 units

Ans. (d) 3 units

8. In the given figure, AB = BC = 10 cm. If AC = 7 cm, then the length of BP is :



(a) 3.5 cm **(b)** 7 cm **(c)** 6.5 cm **(d)** 5 cm

Ans. (c) 6.5 cm

9. Water in a river which is 3 m deep and 40 m wide is flowing at the rate of 2 km/h. How much water will fall into the sea in 2 minutes ?

(a) 800 m^3 **(b)** 4000 m^3 **(c)** 8000 m^3 **(d)** 2000 m^3

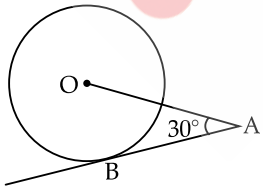
Ans. (c) 8000 m^3

10. If the mean and the median of a data are 12 and 15 respectively, then its mode is:

(a) 13.5 **(b)** 21 **(c)** 6 **(d)** 14

Ans. (b) 21

11. In the given figure, AB is a tangent to the circle centered at O. If OA = 6 cm and $\angle$ OAB = 30°, then the radius of the circle is:



(a) 3 cm **(b)** $3\sqrt{3}$ cm **(c)** 2 cm **(d)** $\sqrt{3}$ cm

Ans. (a) 3 cm

12. $\left(\frac{2\tan 30°}{1+\tan^2 30°}\right)$ is equal to :

(a) sin 60° **(b)** cos 60° **(c)** tan 60° **(d)** sin 30°

Ans. (a) sin 60°

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13. In Δ ABC and Δ DEF, $\frac{AB}{DE} = \frac{BC}{FD}$. Which of the following makes the two triangles similar ?

(a) $\angle A = \angle D$ **(b)** $\angle B = \angle D$ **(c)** $\angle B = \angle E$ **(d)** $\angle A = \angle F$

Ans. (b) $\angle B = \angle D$

14. The 11th term from the end of the A.P. : 10, 7, 4, ... ,– 62 is :

(a) 25 **(b)** 16 **(c)** – 32 **(d)** 0

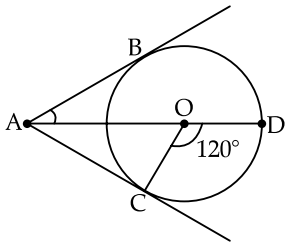
Ans. (c) – 32

15. Two coins are tossed together. The probability of getting at least one tail is :

(a) $\frac{1}{4}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{3}{4}$ **(d)** 1

Ans. (c) $\frac{3}{4}$

16. In the given figure, AC and AB are tangents to a circle centered at O. If $\angle COD = 120°$, then $\angle BAO$ is equal to:



(a) 30° **(b)** 60° **(c)** 45° **(d)** 90°

Ans. (a) 30°

17. Which of the following numbers cannot be the probability of happening of an event?

(a) 0 **(b)** $\frac{7}{0.01}$ **(c)** 0.07 **(d)** $\frac{0.07}{3}$

Ans. (b) $\frac{7}{0.01}$

18. If every term of the statistical data consisting of n terms is decreased by 2, then the mean of the data :

(a) decreases by 2 **(b)** remains unchanged

(c) decreases by $2n$ **(d)** decreases by 1

Ans. (a) decreases by 2

Questions number **19** and **20** are Assertion and Reason based questions carrying 1 mark each. Two statements are given, one labelled as Assertion (a) and the other is labelled as Reason (R). Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below.

(a) Both Assertion (a) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (a).

(b) Both Assertion (a) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (a).

(c) Assertion (a) is true, but Reason (R) is false.

(d) Assertion (a) is false, but Reason (R) is true.

19. Assertion (a): If the points A(4, 3) and B(x, 5) lie on a circle with centre O(2, 3), then the value of x is 2.
Reason (R): Centre of a circle is the mid-point of each chord of the circle.

Ans (c) Assertion (A) is true, but Reason (R) is false.

20. Assertion (a): The number 5^n cannot end with the digit 0, where n is a natural number.
Reason (R): Prime factorisation of 5 has only two factors, 1 and 5.

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Ans. (c) Assertion (A) is true, but Reason (R) is false.
[only 1 prime factor = 5 ∵ prime factorisation of a prime number is the number itself]

SECTION – B

This section comprises very short answer (VSA) type questions of 2 marks each.

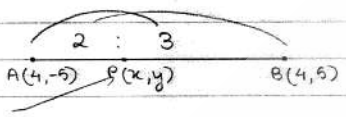
21. (a) The line segment joining the points A(4,–5) and B(4, 5) is divided by the point P such that AP : AB = 2 : 5. Find the coordinates of P.

OR

(b) Point P(x, y) is equidistant from points A(5, 1) and B(1, 5). Prove that $x = y$.

Ans. A(4, −5) and B(4, 5)

(a) Let the coordinates of point P which divides AB such that AP : AB = 2 : 5 be P(x, y).



$\frac{AP}{AB} = \frac{2}{5}$

$\Rightarrow \frac{AP + PB}{AP} = \frac{5}{2}$

$\Rightarrow 2PB = 3AP \Rightarrow AP : PB = 2 : 3$

P divides AB internally in the ratio of 2 : 3

By section formula,

$P(x, y) = P\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$

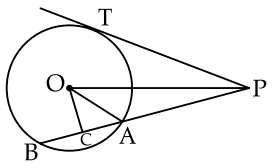
$\Rightarrow P(x, y) = P\left(\frac{2\times4 + 3\times4}{2+3}, \frac{2\times5 + 3\times(-5)}{2+3}\right)$

$\Rightarrow P(x, y) = P\left(\frac{20}{5}, \frac{-5}{5}\right) = P(4, -1)$

∴ Coordinates of P are P(4, −1).

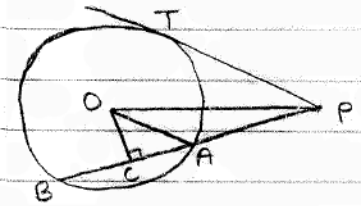
22. In the given figure, PT is a tangent to the circle centered at O. OC is perpendicular to chord AB. Prove that PA.

$PB = PC^2 - AC^2$.



Ans. Given: Circle with centre O
PT is a tangent
OC ⊥ AB

To Prove: $PA \cdot PB = PC^2 - AC^2$



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Perpendicular from centre to chord bisects it
$\Rightarrow$ C is midpoint of AB ($\because \angle OCA = 90°$, given)
$\Rightarrow BC = CA = \frac{AB}{2}$

$\Rightarrow AB = 2AC$

By Pythagoras Theorem,
In right $\triangle OCA$, $OA^2 = OC^2 + AC^2$
In right $\triangle OCP$, $OP^2 = OC^2 + PC^2$

LHS = PA · PB
= PA × (PA + AB)
= PA^2 + PA · AB
= PA^2 + PA · 2AC (from ①)
= $(PA)^2 + 2(PA)(AC)$
= $(PA + AC)^2 - AC^2$ [$(a+b)^2 = a^2 + 2ab + b^2$]
= $PC^2 - AC^2$ = RHS
Hence, proved

23. Using prime factorisation, find HCF and LCM of 96 and 120.

Ans. First number = 96
Second number = 120

$96 = 2^5 \times 3$
$120 = 2^3 \times 3 \times 5$
HCF (96, 120) = $2^3 \times 3$
= 8 × 3
= 24

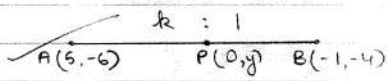
2 | 96	2 | 120
2 | 48	2 | 60
2 | 24	2 | 30
2 | 12	3 | 15
2 | 6	5
3	

LCM (96, 120) = $2^5 \times 3 \times 5$
= 32 × 3 × 5
= 480

$\therefore$ HCF (96, 120) = 24 and LCM (96, 120) = 480

24. Find the ratio in which y-axis divides the line segment joining the points (5, – 6) and (– 1, – 4).

Ans. Points are A(5, -6) and B(-1, -4)
Let the point where y-axis (x = 0) intersects AB be P(0, y)



Let the ratio in which P(0, y) divides AB be k : 1

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By section formula,

$$P(x, y) = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$$

$$P(0, y) = \left(\frac{-k+5}{k+1}, \frac{-4k+(-6)}{k+1}\right)$$

$0 = \frac{-k+5}{k+1}$ and $y = \frac{-4k-6}{k+1}$

$0 = -k + 5$

$k = 5$

$k : 1 = 5 : 1$

∴ Ratio is **5 : 1**

25. (a) If $a \cos\theta + b \sin\theta = m$ and $a \sin\theta - b \cos\theta = n$, then prove that $a^2 + b^2 = m^2 + n^2$.

OR

(b) Prove that :

$$\sqrt{\frac{\sec A - 1}{\sec A + 1}} + \sqrt{\frac{\sec A + 1}{\sec A - 1}} = 2\text{coses } A$$

Ans.

(a) $a\cos\theta + b\sin\theta = m$

$a\sin\theta - b\cos\theta = n$

To prove: $a^2 + b^2 = m^2 + n^2$

$a\cos\theta + b\sin\theta = m$

Squaring both sides,

$a^2\cos^2\theta + b^2\sin^2\theta + 2ab\sin\theta\cos\theta = m^2$ $\left[(a+b)^2 = a^2 + b^2 + 2ab\right]$

$a\sin\theta - b\cos\theta = n$

Squaring both sides,

$a^2\sin^2\theta + b^2\cos^2\theta - 2ab\sin\theta\cos\theta = n^2$ $\left[(a-b)^2 = a^2 + b^2 - 2ab\right]$

Adding ① and ②,

$(a^2\cos^2\theta + a^2\sin^2\theta) + (b^2\sin^2\theta + b^2\cos^2\theta) + 2ab\sin\theta\cos\theta - 2ab\sin\theta\cos\theta = m^2 + n^2$

$\Rightarrow a^2(\sin^2\theta + \cos^2\theta) + b^2(\sin^2\theta + \cos^2\theta) = m^2 + n^2$

$\because \sin^2\theta + \cos^2\theta = 1$

$\Rightarrow a^2 + b^2 = m^2 + n^2$

LHS = RHS

Hence, proved

SECTION – C

This section comprises short answer (SA) type questions of 3 marks each.

26. (a) Prove that $\sqrt{3}$ is an irrational number.

OR

(b) The traffic lights at three different road crossings change after every 48 seconds, 72 seconds and 108 seconds respectively. If they change simultaneously at 7 a.m., at what time will they change together next ?

Ans.

(a) Let us assume, to the contrary, that $\sqrt{3}$ is rational.

$\Rightarrow \sqrt{3} = \frac{p}{q}$ where $q \neq 0$, p and q are coprime positive integers

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Squaring both sides,

$\Rightarrow 3 = \frac{p^2}{q^2}$

$\Rightarrow p^2 = 3q^2$

$\Rightarrow$ 3 divides p^2

$\Rightarrow$ 3 divides p ($\because$ 3 is prime)

So, let $p = 3m$

Substituting in

$(3m)^2 = 3q^2$

$\Rightarrow 9m^2 = 3q^2$

$\Rightarrow q^2 = 3m^2$

$\Rightarrow$ 3 divides q^2

$\Rightarrow$ 3 divides q

From (3) and (4),

3 divides both p and q

But p and q are coprime, i.e. HCF$(p, q) = 1$ (Using (1))

which is a contradiction

$\therefore$ our supposition is wrong

$\therefore \sqrt{3}$ must be irrational.

Hence, proved

27. If p^{th} term of an A.P. is q and q^{th} term is p, then prove that its nth term is $(p + q - n)$.

Ans. Let the first term of AP be a

and common difference be d

AP $a, a+d, a+2d, \ldots$

$a_n = a + (n-1)d$

$a_p = a + (p-1)d = q$ ...(1)

$\Rightarrow a + pd - d = q$

$a_q = a + (q-1)d = p$ ...(2)

$\Rightarrow a + qd - d = p$

Subtracting (1) from (2),

$a + qd - d = p$

$-\quad a + pd - d = q$

$(q-p)d = p-q$

$\Rightarrow d = -1$

Substituting in (1),

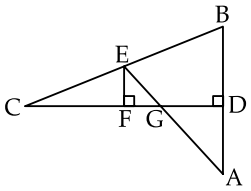
$a + p(-1) - (-1) = q$

$\Rightarrow a + 1 = p + q$

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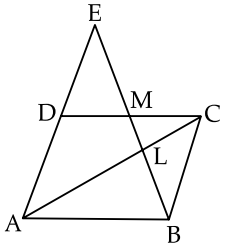
$a_n = a + (n-1)d$
$= a + (n-1)(-1)$
$= a - n + 1$
$= (a+1) - n$
$= p + q - n$ (Substituting from (3))
LHS = RHS
Hence, proved

28. (a) In the given figure, CD is the perpendicular bisector of AB. EF is perpendicular to CD. AE intersects CD at G. Prove that $\frac{CF}{CD} = \frac{FG}{DG}$.



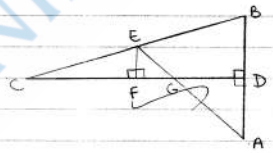
OR

(b) In the given figure, ABCD is a parallelogram. BE bisects CD at M and intersects AC at L. Prove that EL = 2BL.



Ans.

(a) Given: CD is ⊥ bisector of AB
EF ⊥ CD
AE intersects CD at G



To Prove: $\frac{CF}{CD} = \frac{FG}{DG}$

Proof: In $\triangle EFG$ and $\triangle ADG$,

$\angle EFG = \angle ADG = 90°$	(given, linear pair with $\angle EFC$ and $\angle BDG$)
$\angle EGF = \angle AGD$	(vertically opposite angles)
$\therefore \triangle EFG \sim \triangle ADG$	(by AA similarity criterion)
$\Rightarrow \frac{EF}{AD} = \frac{FG}{DG}$	(cpst)

In $\triangle ECF$ and $\triangle BCD$,

$\angle EFC = \angle BDC = 90°$	(given)
$\angle ECF = \angle BCD$	(common angle)
$\therefore \triangle ECF \sim \triangle BCD$	(by AA similarity criterion)
$\Rightarrow \frac{EF}{BD} = \frac{CF}{CD}$	(cpst)

But $AD = BD$ ($\because$ CD bisects AB)

$\Rightarrow \frac{EF}{AD} = \frac{EF}{BD} = \frac{CF}{CD}$

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From item ① and ②,

$$\frac{EF}{AD} = \frac{CF}{CD} = \frac{FG}{DG}$$

$$\therefore \frac{FG}{CD}\,\frac{CF}{DG} = \frac{FG}{DG}$$

Hence, proved

29. Two people are 16 km apart on a straight road. They start walking at the same time. If they walk towards each other with different speeds, they will meet in 2 hours. Had they walked in the same direction with same speeds as before, they would have met in 8 hours. Find their walking speeds.

Ans. Let the speed of person 1 be x km/h and speed of person 2 be y km/h $(x > y)$

distance = speed × time

Given distance = 16 km

Case 1 : Towards each other

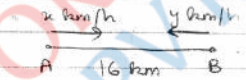
Time = 2 h

Speed = $(x+y)$ km/h

Distance = $2(x+y)$ km

$\Rightarrow 16 = 2x + 2y$

$\Rightarrow x + y = 8$ ... ①



Case 2 Same direction

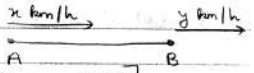
Time = 8 h

Speed = $(x-y)$ km/h

Distance (only AB) = $8(x-y)$ [$16 + 8y = 8x$]

$\Rightarrow 16 = 8x - 8y$

$\Rightarrow x - y = 2$ ... ②



Adding ① and ②,

$x + y = 8$

$+ \quad x - y = 2$

$2x = 10$

$\Rightarrow x = 5$

Substituting in ①,

$y = 3$ km

$\therefore$ Speed of person 1 = 5 km/h

Speed of person 2 = 3 km/h

30. Prove that :

$$\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta\,\mathrm{cosec}\,\theta$$

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Ans. $\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta\,\text{cosec}\,\theta$

$$\text{LHS} = \frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta}$$

$$= \frac{\tan\theta}{1-\frac{1}{\tan\theta}} + \frac{\frac{1}{\tan\theta}}{1-\tan\theta}$$

$$= \frac{\tan^2\theta}{\tan\theta - 1} + \frac{1}{\tan\theta(1-\tan\theta)}$$

$$= \frac{\tan^2\theta}{\tan\theta - 1} - \frac{1}{\tan\theta(\tan\theta - 1)}$$

$$= \frac{\tan^3\theta - 1}{(\tan\theta - 1)\tan\theta}$$

$$= \frac{(\tan\theta - 1)(\tan^2\theta + 1 + \tan\theta)}{\tan\theta(\tan\theta - 1)} \qquad \left[(a-b)(a^2+b^2+ab) = a^3 - b^3\right]$$

$$= \frac{1 + \tan^2\theta + \tan\theta}{\tan\theta}$$

$$= \frac{\tan\theta + \sec^2\theta}{\tan\theta} \qquad (1 + \tan^2\theta = \sec^2\theta)$$

$$= \frac{\tan\theta}{\tan\theta} + \frac{\sec^2\theta}{\tan\theta}$$

$$= 1 + \frac{1}{\cos^2\theta} \times \frac{\cos\theta}{\sin\theta} \qquad \left(\sec\theta = \frac{1}{\cos\theta},\ \tan\theta = \frac{\sin\theta}{\cos\theta}\right)$$

$$= 1 + \frac{1}{\cos\theta\sin\theta}$$

$$= 1 + \sec\theta\,\text{cosec}\,\theta$$

$$= \text{RHS}$$

Hence, proved

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31. Find the mean of the following frequency distribution:

Classes	25-30	30-35	35-40	40-45	45-50	50-55	55-60
Frequency	14	22	16	6	5	3	4

Ans.

Classes	Frequency (f_i)	Midpoints (x_i)	$u_i = \frac{x_i - A}{h}$	$f_i u_i$
25 - 30	14	27.5	-3	-42
30 - 35	22	32.5	-2	-44
35 - 40	16	37.5	-1	-16
40 - 45	6	A (42.5)	0	0
45 - 50	5	47.5	1	5
50 - 55	3	52.5	2	6
55 - 60	4	57.5	3	12
	$\Sigma f_i = 70$			$\Sigma f_i u_i = -79$

Mean = $\bar{x} = A + \frac{h\Sigma f_i u_i}{\Sigma f_i}$

$= 42.5 + \frac{5 \times (-79)}{70}$

$= 42.5 - \frac{395}{70}$

$= 42.5 - 5.642$

$= 36.858 \approx 36.86$

∴ Mean = 36.86 (approx)

SECTION – D

This section comprises long answer (LA) type questions of 5 marks each.

32. One observer estimates the angle of elevation to the basket of a hot air balloon to be 60°, while another observer 100m away estimates the angle of elevation to be 30°. Find :

(a) The height of the basket from the ground.

(b) The distance of the basket from the first observer's eye.

(c) The horizontal distance of the second observer from the basket.

Ans.

Introduction

2 cases are formed

A = hot-air ballon

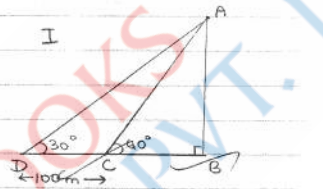
C = first observer

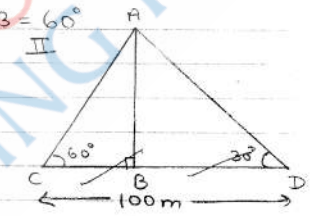
D = second observer

Angle of elevation of A from C = ∠ACB = 60°

Angle of elevation of A from BD = ∠ADB = 30°

CD = 100 m





(a) Let height of basket = AB = h m

Case I : In △ABC,

$\tan 60° = \frac{AB}{BC}$ ⇒ $\sqrt{3} = \frac{h}{BC}$ ⇒ $BC = \frac{h}{\sqrt{3}}$ ①

In △ABD,

$\tan 30° = \frac{AB}{BD}$ ⇒ $\frac{1}{\sqrt{3}} = \frac{h}{BC + 100}$ ⇒ $BC + 100 = h\sqrt{3}$

$BC + 100 = h\sqrt{3}$

⇒ $\frac{h}{\sqrt{3}} + 100 = h\sqrt{3}$

⇒ $100 = h\sqrt{3} - \frac{h\sqrt{3}}{3}$

⇒ $\frac{2h\sqrt{3}}{3} = 100$

⇒ $h = \frac{150}{\sqrt{3}} = 50\sqrt{3}$ m

∴ Height of basket = $\boxed{50\sqrt{3}\text{ m}}$

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Case 2 : In $\triangle ABC$,

$\tan 60^\circ = \frac{AB}{BC}$

$\Rightarrow \sqrt{3} = \frac{h}{BC}$

$\Rightarrow BC = \frac{h}{\sqrt{3}}$

In $\triangle ABD$,

$\tan 30^\circ = \frac{AB}{BD}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{BD}$

$\Rightarrow BD = h\sqrt{3}$

$BC + BD = 100\text{ m}$

$\Rightarrow \frac{h}{\sqrt{3}} + h\sqrt{3} = 100$

$\Rightarrow \frac{4h\sqrt{3}}{3} = 100$

$\Rightarrow h = 25\sqrt{3}$

$\therefore$ Height of basket = $\boxed{25\sqrt{3}\text{ m}}$

(b). Case 1 : Dist of A from C = AC

In $\triangle ABC$,

$\sin 60^\circ = \frac{AB}{AC}$

$\Rightarrow \frac{\sqrt{3}}{2} = \frac{50\sqrt{3}}{AC}$

$\Rightarrow AC = 100\text{ m}$

$\therefore$ Distance of basket from first observer = $\boxed{100\text{ m}}$

Case 2 : Dist. of A from C = AC

In $\triangle ABC$,

$\sin 60^\circ = \frac{AB}{AC}$

$\Rightarrow \frac{\sqrt{3}}{2} = \frac{25\sqrt{3}}{AC}$

$\Rightarrow AC = 50\text{ m}$

$\therefore$ Distance of basket from first observer = $\boxed{50\text{ m}}$

(c) Case 1 : To find - BD

$BD = BC + CD$

$= \frac{h}{\sqrt{3}} + 100$ (from ①)

$= \frac{50\sqrt{3}}{\sqrt{3}} + 100$

$= 50 + 100 = 150$

$\therefore$ Horizontal distance BD = $\boxed{150\text{ m}}$

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Case 2. To find - BD

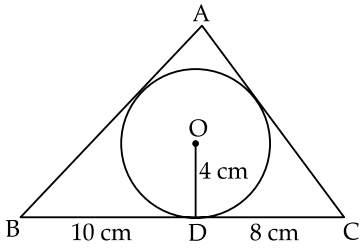
$BD = h\sqrt{3}$ (from ②)

$= 25\sqrt{3} \times \sqrt{3}$

$= 75$

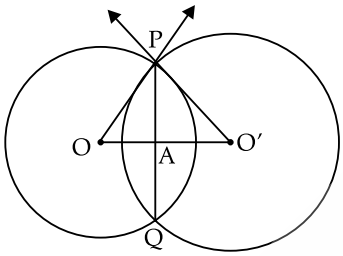
∴ Horizontal distance BD = 75 m

33. (a) A triangle ABC is drawn to circumscribe a circle of radius 4 cm such that the segments BD and DC are of lengths 10 cm and 8 cm respectively. Find the lengths of the sides AB and AC, if it is given that area □ ABC= 90 cm^2.



OR

(b) Two circles with centres O and O' of radii 6 cm and 8 cm, respectively intersect at two points P and Q such that OP and O'P are tangents to the two circles. Find the length of the common chord PQ.



Ans.

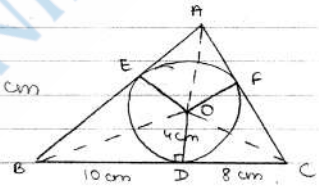
(a) Given: △ABC

Incircle with radius $r = 4$ cm

BD = 10 cm

CD = 8 cm

ar(ABC) = 90 cm^2



To find: Lengths of AB and AC.

Construction: Join OE, OF, OA, OB, OC

OD = OE = OF = r = 4cm (radii)

Tangents from same external point are equal in length.

From pt. A, AE = AF = x (let it be)

From pt. B, BE = BD = 10 cm

From pt C, CD = CF = 8 cm

Area of $\triangle = \frac{1}{2} \times b \times h$

Area of △ABC = Area of △OAB + Area of △OBC + Area of △OCA

$\Rightarrow 90 = \frac{1}{2} \times OE \times AB + \frac{1}{2} \times OD \times BC + \frac{1}{2} \times OF \times AC$

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$+\frac{1}{2} \times x \times (CF + AF)$

$\Rightarrow \quad 90 = \frac{1}{2} \times x \times (x + 10) + \frac{1}{2} \times x \times (10 + 8) + \frac{1}{2} \times x \times (8 + x)$

$\Rightarrow \quad 90 = \frac{1}{2} \times 4 \,(x + 10 + 10 + 8 + 8 + x)$

$\Rightarrow \quad 45 = 2x + 36$

$\Rightarrow \quad 2x = 9$

$\Rightarrow \quad x = 4.5$

$AB = AE + BE = 4.5 + 10 = 14.5 \text{ cm}$

$AC = AF + CF = 4.5 + 8 = 12.5 \text{ cm}$

$\therefore \quad AB = \boxed{14.5 \text{ cm}}$

$AC = \boxed{12.5 \text{ cm}}$

34. (a) A train travels at a certain average speed for a distance of 54 km and then travels a distance of 63 km at an average speed of 6 km/h more than the first speed. If it takes 3 hours to complete the journey, what was its first average speed ?

OR

(b) Two pipes together can fill a tank in $\frac{15}{8}$ hours. The pipe with 8 larger diameter takes 2 hours less than the pipe with smaller diameter to fill the tank separately. Find the time in which each pipe can fill the tank separately.

Ans. (a) Let the original average speed be x km/h.

Original : distance = 54 km

speed = x km/h

time = $\frac{\text{distance}}{\text{speed}} = \frac{54}{x}$ h

New : distance = 63 km

speed = $(x + 6)$ km/h

time = $\frac{\text{distance}}{\text{speed}} = \frac{63}{x+6}$ h

$\frac{54}{x} + \frac{63}{x+6} = 3$

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$\Rightarrow 3\left(\frac{18}{x} + \frac{21}{x+6}\right) = 3$

$\Rightarrow \frac{18x + 108 + 21x}{x^2 + 6x} = 1$

$\Rightarrow 39x + 108 = x^2 + 6x$

$\Rightarrow x^2 - 33x - 108 = 0$

$\Rightarrow x^2 - 36x + 3x - 108 = 0$

$\Rightarrow x(x-36) + 3(x-36) = 0$

$\Rightarrow (x-36)(x+3) = 0$

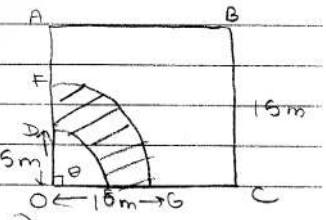
$\Rightarrow x = 36$ or $x = -3$

Speed cannot be negative,
$x = -3$ will be neglected
$\Rightarrow x = 36$

∴ Average speed of team (original) = 36 km/h

35. A horse is tied to a peg at one corner of a square shaped grass field of side 15 m by means of a 5 m long rope. Find the area of that part of the field in which the horse can graze. Also, find the increase in grazing area if length of rope is increased to 10m. (Use $\pi = 3.14$)

Ans.



Square field OABC
Side = s = 15 m

Quadrant
$\angle DOE = \theta = 90^\circ$ (all angles of square = 90°)
Radius = length of rope
$= r = 5$ m

Area grazed by horse = Area of sector DOE

$= \frac{\theta}{360^\circ} \times \pi \times r^2$

$= \frac{90}{360} \times \frac{157}{50} \times 5 \times 5$

$= \frac{3925}{200}$

$= 19.625\ m^2$

157
25
785
314×
3925×3
2 7175
= 1962 5

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New, new radius = new length of arc = R = 10 m

Increase in grazing area = Area of sector FOG − Area of sector DOE

$$= \frac{\theta}{360^\circ} \times \pi \times R^2 - \frac{\theta}{360^\circ} \times \pi \times r^2$$

$$= \frac{90}{360} \times \frac{314}{100} \times (10^2 - 5^2)$$

$$= \frac{90}{360} \times \frac{314}{100} \times 15 \times 5 \quad [a^2 - b^2 = (a+b)(a-b)]$$

$$= \frac{11775}{200}$$

$$= 58.875 \text{ m}^2$$

Original area grazed = 19.625 m^2

Increase in area = 58.875 m^2

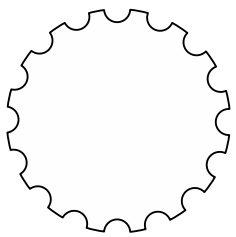
SECTION – E

This section comprises 3 case study based questions of 4 marks each.

Case Study - 1

36. A golf ball is spherical with about 300 - 500 dimples that help increase its velocity while in play. Golf balls are traditionally white but available in colours also. In the given figure, a golf ball has diameter 4.2 cm and the surface has 315 dimples (hemi-spherical) of radius 2 mm.





Based on the above, answer the following questions :

(i) Find the surface area of one such dimple. **1**

(ii) Find the volume of the material dug out to make one dimple. **1**

(iii)(a) Find the total surface area exposed to the surroundings. **2**

OR

(iii)(b) Find the volume of the golf ball. **2**

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Ans.

Golf ball = Sphere

Radius = $R = \frac{4.2}{2} = 2.1$ cm

Dimple = Hemisphere

Radius = $r = 2$ mm $= 0.2$ cm

(i) SA of 1 dimple = CSA of hemisphere

$= 2\pi r^2$

$= 2 \times \frac{22}{7} \times \frac{2}{10} \times \frac{2}{10}$

$= \frac{176}{700}$ cm^2

$= 0.2514$ cm^2

(ii) Vol to make 1 dimple = Vol of hemisphere

$= \frac{2\pi r^3}{3}$

$= \frac{2}{3} \times \frac{22}{7} \times \frac{2}{10} \times \frac{2}{10} \times \frac{2}{10}$

$= \frac{352}{21000}$

$= 0.01676$ cm^3

(iii) (b.) Vol of golf ball = Vol of sphere − Vol of 315 hemispheres

$= \frac{4\pi R^3}{3} - \frac{2\pi r^3 \times 315}{3}$

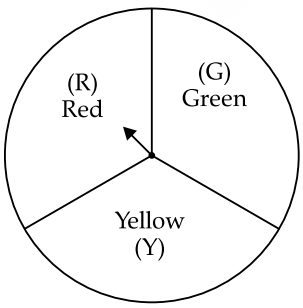
$= \frac{4 \times 22}{3} \times \frac{21}{10} \times \frac{21}{10} \times \frac{21}{10} - \frac{2 \times 22}{3} \times \frac{2}{7} \times \frac{2}{10} \times \frac{2}{10} \times \frac{315}{10}$

$= 38.808 - 5.28$

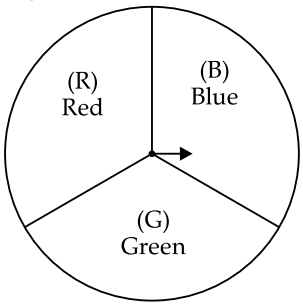
$= 33.528$ cm^3

Case Study - 2

37. A middle school decided to run the following spinner game as a fund-raiser on Christmas Carnival.



I



II

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Making Purple : Spin each spinner once. Blue and red make purple. So, if one spinner shows Red (R) and another Blue (b), then you 'win'. One such outcome is written as 'RB'.

Based on the above, answer the following questions :

(i) List all possible outcomes of the game. **1**

(ii) Find the probability of 'Making Purple'. **1**

(iii)(a) For each win, a participant gets ₹ 10, but if he/she loses, he/she has to pay ₹ 5 to the school. If 99 participants played, calculate how much fund could the school have collected. **2**

OR

(iii)(b) If the same amount of ₹ 5 has been decided for winning or losing the game, then how much fund had been collected by school? (Number of participants = 99) **2**

Ans.

(i) Spinner I – Spinner II

Red (R) – Red (R) } RR,
Red (R) – Blue (B) RB,
Red (R) – Green (G) RG,
Green (G) – Red (R) GR,
Green (G) – Blue (B) GB,
Green (G) – Green (G) GG,
Yellow (Y) – Red (R) YR,
Yellow (Y) – Blue (B) YB,
Yellow (Y) – Green (G) YG }

Total no. of outcomes = 9

(ii) X = {RB}

Favourable outcomes = 1

P(making purple) = No. of favourable outcomes / Total no. of outcomes = 1/9

(iii)(a) No. of participants = 99

Winning No. = $\frac{1}{9} \times 99 = 11$

Amount = ₹(−10) (school has to pay)

Value = ₹(−110) [loss]

Less No. = 99 − 11 = 88

Amount = ₹ 5 (pay to school)

Value = ₹ 440 [gain]

Net = + 440 − 110
= + 330

∴ So, the school most likely collected Rs. 330

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Case Study - 3

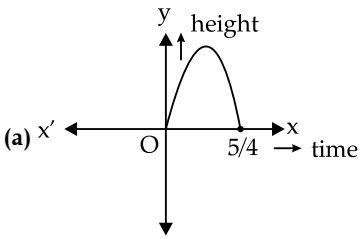
38. In a pool at an aquarium, a dolphin jumps out of the water travelling at 20 cm per second. Its height above water level after t seconds is given by $h = 20t - 16t^2$.

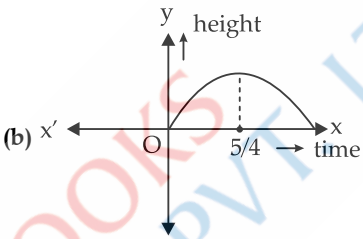


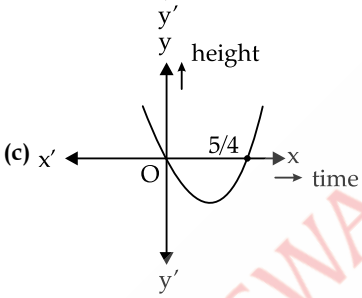
Based on the above, answer the following questions :

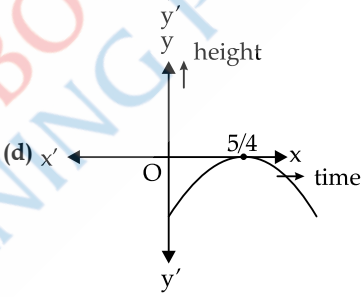
(i) Find zeroes of polynomial $p(t) = 20t - 16t^2$. **1**

(ii) Which of the following types of graph represents $p(t)$? **1**

(a)


(b)


(c)


(d)


Ans.

(iii)(a) What would be the value of h at $t = \frac{3}{2}$? Interpret the result. **2**

OR

(b) How much distance has the dolphin covered before hitting the water level again ? 2

(i) $p(t) = 20t - 16t^2$

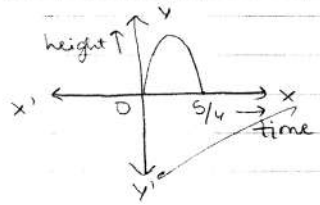
$20t - 16t^2 = 0$

$-4t(4t - 5) = 0$

$t = 0$ or $t = \frac{5}{4}$

∴ Zeroes of $p(t) = 20t - 16t^2$ are 0 and $\frac{5}{4}$

(ii) (a)



→ opens downwards ($a < 0$)

→ intersects x-axis at $(0, 0)$ and $\left(\frac{5}{4}, 0\right)$

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(iii) Water level is hit when $h = 0$

(iv) $h = 20t - 16t^2$

$\Rightarrow 0 = -4t(4t - 5)$

$\therefore t = 0$ or $t = \frac{5}{4}$

$\because$ Dolphin has started at $t = 0$

$\therefore$ at $t = \frac{5}{4}$, dolphin reaches water level again.

distance covered = speed × time

$= 20 \text{ cm/s} \times \frac{5}{4} \text{ s}$ 2

$= \boxed{25 \text{ cm}}$

■■

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